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Assembly for securely coupling a tool on a working arm of a public works machine with improved operation

Abstract

Quick coupler for coupling a tool on a working arm of a public works machine, with two connection components equipping the tool, two coupling components for coupling with the connection components, a security component mounted for free rotation, at least one locking component, a translationally movable release component adapted to allow the security component to rotate. The release component with at least one translationally movable part adapted to be in contact with at least one surface, the shape of which is adapted to that of the security component, and a security component with at least one surface adapted to be in sliding contact with at least one complementary shaped surface of a connection component, the sliding contact inducing a rotation of the security component when the connection component moves out of its accommodation housing defined by the first coupling component.

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Background/Summary

(1) The present invention relates to an assembly for securely coupling a tool on a working arm of a public works machine with improved operation.

DESCRIPTION OF RELATED ART

(2) The expression public works machine denotes any machine, of the self-propelled vehicle type or a stationary machine, positioned on land, on a vessel, on a pontoon or on a railway vehicle, equipped with at least one arm, at the end of which a tool is detachably mounted, with said tool being of the shovel, bucket, gripper, chisel, hammer, basket, lifting hook type or any other tool used within the context of heavy on-site, civil engineering, transport, agricultural, mining and quarrying or handling work.

(3) The arms of a public works machine can accommodate various tools, both by their function and by their size, in order to undertake different jobs. These tools can move along one or more axes

relative to the end of the arm of the vehicle. A tool is coupled on the end of a working arm of a public works machine using a coupling assembly that comprises a plurality of parts and is known as a quick coupler assembly, more easily denoted using the expression “quick coupler,” which will preferably be used hereafter.

(4) Such a quick coupler allows the tool to be securely coupled to the working arm without human intervention. Many systems also allow the tool to be securely uncoupled from the working arm. The tools used on public works machines comprise two parallel connection shafts, allowing coupling with the quick coupler. Said quick coupler is provided with components for coupling and locking the shafts of the tool on the coupler. A coupler is known from EP-B-1 318 242 for which the translation movement of a jack, controlled by the user from the operating cab of the public works machine, ensures that the two coupling components of the quick coupler are locked and unlocked. WO-A-2020 128 075 describes a locking device in which a jack translationally activates hook-shaped coupling components adapted to retain the shafts of the tool. A blocking flap mounted for free rotation securely retains a coupling component on a shaft. The unlocking movement of the jack is accompanied by the compression of a spring, with said spring driving a part coming into abutment on the flap. Said flap is rotated to a position where it no longer prevents the shaft from exiting the retention component. EP-B-2 466 015 discloses a solution where two hooks form the shaft coupling components. A hook is directly connected by a pivoting link to the rod of a jack. The other hook is connected by a pivoting link to the rod of the jack by an arm. The movement of the rod of the jack successively induces the movements of the two hooks. EP-A-2 987 916 describes an activation component in the shape of a tuning fork that slides and pivots a component for blocking one of the shafts in position in the coupler. The disengagement of the shafts of the quick coupler may or may not be allowed depending on the position of the tuning fork-shaped component. The translation movement of the activation component, in one direction or the other, is subject to the action of a jack. The solution marketed by DROMONE ENGINEERING comprises a flap for locking a component for coupling the quick coupler with a shaft. The coupling component is moved by the action of a jack, with the locking flap pivoting into the active position under the effect of gravity. This maneuver requires tilting of the tool at the end of the arm. In this position, the other shaft is also locked by a coupling component that is moved by a jack. Unlocking the coupling components also requires tilting of the tool at the end of the arm so that the movement of the jack induces the movement of a lug acting on the flap. WO-A-2018/056841 relates to a quick coupler for coupling a tool. In a first example, the coupler comprises a fixed jaw, configured as a U-shape, and a second jaw mounted on a slide that can move under the action of a hydraulic jack. A spring pushes the slide away from one of the jaws. The coupler comprises a locking component allowing the connection component to be blocked and being configured in the shape of an L and pivotably mounted. A branch of the locking component forms an activation element. A finger defining an unlocking element is mounted on the slide. It is moved when the jack translationally moves. In this case, the movement of the unlocking components, as well as the movement of one of the jaws, is obtained by hydraulic means.

(5) Such solutions require action by the user at least for resetting the device, i.e. returning the device to a position where it is ready to ensure set-up and coupling with a tool. In addition, these solutions are complex to implement and maintain.

(6) Such solutions currently comply with the existing standards, in particular, European standards EN474-1/A3, EN474-1:2006+A5, as well as international standard ISO 13031 (2016) and machine directive: 2006/42/EC. With this being the case, if the operator of the public works machine performs movements with the arm of the vehicle, typically pendulum movements, without having correctly engaged the tool and/or locked the quick coupler, there is a risk of uncontrolled swinging of the tool around one of the connection shafts or of the tool breaking free, and therefore falling to the ground. This risk is even more marked as the coupling and uncoupling kinematics of certain solutions induce movements of the tool and/or a given position of said tool during its coupling

and/or uncoupling with the quick coupler.

(7) Currently, authorities wish to increase the level of security relating to the connection of tools with a quick coupler, as well as the information given to the user with respect to the state of the connection. In particular, the question of the tool potentially falling if the quick coupler is not correctly locked with the tool must be resolved. Therefore, a need exists for a coupling assembly between a tool and an arm of a public works machine, the installation of which tool is simple, secure, with optimized manufacturing and maintenance, while offering maximum assurance against any case of the tool potentially falling if the coupling between the tool and the quick coupler is not implemented correctly.

(8) The invention proposes a response to this need by offering an assembly for coupling a tool on a working arm of a public works machine that is secure, easy to implement and guarantees increased safety by avoiding any risk of the tool unhooking as long as the coupling system is not in the unlocked position, while limiting the number of parts and being adaptable to various types of quick couplers and tools.

BRIEF SUMMARY OF THE INVENTION

(9) To this end, the subject matter of the invention is an assembly for coupling a tool on a working arm of a public works machine, comprising a secure connection device, called quick coupler, two parallel connection components equipping the tool, at least one of the connection components being a cylinder with a circular base, two coupling components with the connection components equipping the secure connection device, said coupling components defining housings for accommodating the connection components, a security component mounted to freely rotate between a position, called disengaged position, in which it allows a first connection component equipping the tool to move relative to a first coupling component, and a position, called engaged position, in which it secures the retention of the first connection component by the first coupling component, at least one locking component that is movable between an inactive position, in which the retention by a second coupling component equipping the secure connection device of the second connection component is not locked, and an active position, in which it locks the retention by the second coupling component of the second connection component, a release component that is translationally movable and is adapted to allow the security component to rotate from the disengaged position to the engaged position, the release component comprising at least one translationally movable part that is fixed on the locking component, said movable part of the release component being adapted to be in contact with at least one surface of suitable shape secured to the security component and said security component comprising at least one surface adapted to be in sliding contact with at least one surface, the shape of which is adapted to the first connection component, said sliding contact inducing a rotation of the security component when the first connection component moves out of its accommodation housing defined by the first coupling component, characterized in that the rotation of the security component when the first connection component moves out of its accommodation housing defined by the first coupling component occurs in the counterclockwise direction, toward an edge of the lower part of the connection device, to a position in which the release component allows the security component to return to a position, called reset position, in which the security component is remote from the first connection component, without any coupling between the tool and the connection device, at least one surface of part of the security component being in a position, called engaged position, corresponding to a position where the engagement of said surface in the housing defined by the first coupling component is maximal, said housing being empty.

(10) Thus, by virtue of the invention, the number of parts is limited, and therefore as is the risk of a malfunction, while facilitating maintenance and ensuring optimum safety, with accidental dropping of the tool during locking no longer being possible.

(11) According to advantageous but non-compulsory aspects of the invention, such a coupling assembly can comprise one or more of the following features:

- (12) The security component comprises a round relief or flap, the radius of curvature of which is adapted to the radius of the cylinder with a circular base forming the first connection component, said relief comprising at least three flat and curved surface segments adapted to be successively at least facing at least part of the surface of the connection component.
 - (13) A segment of the round relief is permanently at least facing part of the surface of the first connection component when said component is in its accommodation housing defined by the first coupling component.
 - (14) The security component is retained in the engaged position by a return component.
 - (15) The release component comprises a main body receiving a finger that is translationally movable in the body.
 - (16) A return component permanently retains the finger outside the body of the release component.
 - (17) A return component for positioning the body retains said body in a position in which the finger is at least permanently facing the security component.
 - (18) The finger comprises at least one zone in contact with the security component and located at a free end of the finger when the retention of the second connection component by the second coupling component is not locked.
 - (19) The finger comprises a notch adapted to receive a pawl that is free to rotate relative to the security component.
 - (20) The pawl is secured to the security component, with a pin for rotating the pawl being secured to a wall of the quick coupler.
 - (21) The end of the finger extends beyond the quick coupler when the security component is in the disengaged position, forming a visual signal for the user.
 - (22) The locking component comprises a lug configured as a hook that is translationally movable in slides.
 - (23) The slides are detachably mounted in the quick coupler.
 - (24) The invention also relates to a public works machine provided with at least one working arm equipped with a coupling assembly for coupling a tool on the working arm in accordance with any one of the preceding features.
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Description

BRIEF DESCRIPTION OF THE VIEW OF THE DRAWINGS

- (1) The invention will be better understood, and further advantages thereof will become more clearly apparent, from reading the following description, which is provided solely by way of a non-limiting example and refers to the accompanying drawings, in which:
- (2) FIG. 1 is a partial side view of an arm of a public works machine equipped with a tool connected to the arm by a coupling assembly according to one embodiment of the invention;
- (3) FIG. 2 is a larger scale side view of the tool and of the secure connection device forming the coupling assembly of FIG. 1, alone and in the uncoupled position;
- (4) FIG. 2A is a larger scale view of the detail surrounded with broken lines in FIG. 2;
- (5) FIG. 3 is a larger scale partial view of the elements of FIG. 2, in a close position, with the first coupling component being ready to receive the first connection component of the tool;
- (6) FIG. 4 is a view similar to FIG. 3, with the first coupling component being in the process of receiving the first connection component of the tool, with the security component being in a position allowing engagement;
- (7) FIG. 5 is a view similar to FIG. 4, with the first connection and coupling components being mutually engaged, the security component being in position for securing the retention of the connection component in the accommodation housing defined by the coupling component;
- (8) FIG. 6 is a view similar to FIG. 5, with a second connection component of the tool and a second

- coupling component of the secure connection device being in the process of coupling;
- (9) FIG. 7 is a view similar to FIG. 6, with the two connection components of the tool being securely retained by the two coupling components of the secure connection device, the locking and security components being in an active position;
- (10) FIG. 8 is a view similar to FIG. 7, illustrating the beginning of the uncoupling of the tool and the quick coupler, with the locking component being in the process of transitioning to the inactive position;
- (11) FIG. 9 is a view similar to FIG. 8, with the security component having pivoted, the release component being at the end of its travel and the locking component being in the inactive position, the second connection and coupling components being uncoupled;
- (12) FIG. 10 is a view similar to FIG. 9, with the uncoupling of the first connection and coupling components being almost total, the assembly being in a configuration similar to that illustrated in FIG. 4;
- (13) FIG. 11 is a partial perspective view, to another scale, of part of the secure connection device or quick coupler of FIGS. 1 to 10 illustrating the various components;
- (14) FIG. 12 is a larger scale side view of the security component illustrated in FIGS. 1 to 10 alone; and
- (15) FIG. 13 and FIG. 14 are views similar to FIGS. 7 and 10, respectively, of the invention according to another embodiment.

DETAILED DESCRIPTION OF THE INVENTION

(16) FIG. 1 illustrates an arm 1 of a public works machine, with said machine being known per se and not being illustrated. In this case, it is a machine of the mechanical shovel type. It is easily understood that such a public works machine can be of another type, for example, an excavator, a backhoe, a dredger, a mower, a loading machine or more generally any self-propelled or non-self-propelled land, railway or maritime machine equipped with at least one working arm equipped with at least one working tool. In this case, the working tool 2 is a bucket. As an alternative embodiment, it can be another tool, for example, a blade, a rock breaker, a pestle, a basket, a crushing/clearing head or the like. The free end 3 of the arm 1 is equipped with an articulated lever 4. The end 3 and a free end 5 of the lever 4 are each provided with a rotation shaft 6. In this case, the shafts 6 are identical. As an alternative embodiment, they are different. These shafts 6 maintain and allow movement, following the double arrow F, of a secure connection device 7 for connecting the tool 2 on the arm 1. Such a device is frequently denoted using the expression “quick coupler,” which will also be used hereafter. To ensure that the quick coupler 7 is retained on the rotation shafts 6, said shafts are inserted into complementary shaped orifices 8 provided on part of the quick coupler 7. Depending on the configuration of the end 3 of the arm 1, the lever 4, the number and/or the position of the rotation shafts 6, the quick coupler 7, and therefore de facto the tool 2, can be moved in one or more planes, given that the quick coupler can allow, depending on its configuration, the tool to rotate about a vertical axis. Thus, insofar as the configuration of the connection components of the tool allows, the tool 2 can be oriented in order to work toward the machine or away from the machine.

(17) FIG. 2 is a larger scale representation of the tool 2 and the quick coupler 7 of FIG. 1, in the uncoupled position, i.e. before the tool 2 is mounted on the arm 1 or once the tool 2 is unhooked from the arm 1. In such a position, there is no connection, not even a partial connection, between the tool 2 and the arm 1. The secure connection device or quick coupler 7 is a set of parts, made of metal or made of one or more material(s) resistant to shocks and environmental stresses. As shown in FIGS. 1 to 14, the device 7 generally assumes the shape of a rectangular parallelepiped. It comprises a part 9 for fixing onto the arm 1. This part 9 forms the upper part of the coupler 7, with reference to FIG. 2. In addition to the orifices 8 for accommodating the rotation shafts 6, it comprises, in the illustrated embodiment, a lifting hook 10. The hook 10 allows lifting operations to be carried out using a sling. As an alternative embodiment, the part 9 is devoid of a hook 10 or

the like. It is understood that the lifting hook **10**, when present, can assume a shape and/or dimensions other than those illustrated in FIG. 2. The connection between the part **9** and the arm **1**, even if it is detachable, generally remains permanently in place, so that the arm **1** is permanently equipped with the device **7**. Thus, no particular maneuver when moving the public works machine is necessary, with the arm **1** always being ready to accommodate a tool **2**. The device **7** can also be fitted as standard on the arm **1** of a public works machine or even retrofitted on the arm **1** of a public works machine, provided that the orifices **8** are adapted in terms of numbers, dimensions and positions to the rotation shafts **6** equipping the arm **1**. To this end, according to another embodiment, not shown, the part **9** is detachable in order to be easily changed to adapt to the configuration of the arm **1**.

(18) The lower part **11** of the device **7**, with reference to FIG. 2, is dedicated to the secure and detachable connection between the device **7**, and therefore de facto the arm **1**, and the tool **2**. To this end, the tool **2** comprises at least two parallel connection components **12A** and **12B**. In the example illustrated in the various figures, there are two cylindrical metal shafts with a circular base of the same diameter and length. As an alternative embodiment, they are not identical in terms of dimensions and/or geometric shape. In all cases, at least one of the connection components is a shaft or a cylinder with a circular base. The connection components or shafts **12A** and **12B** are located on one end **200** of the tool **2**. They are positioned and dimensioned to provide not only a secure connection of the tool **2** on the device **7**, but also to provide easy and effective maneuvering of the tool **2**. The parallel shafts or connection components **12A**, **12B** are oriented in a direction **D** that is substantially orthogonal to a longitudinal axis of the arm **1** when the tool **2** is mounted on the arm **1**. By convention, with reference to their position illustrated in FIG. 1, relative to the free end **3** of the arm **1** when the tool **2** is mounted on the quick coupler **7**, it is referred to as being the rear connection component, therefore the furthest from the arm **1** in FIG. 1, the connection component or shaft **12B** and as being the connection component before the connection component or shaft **12A**. It is understood that, when the tool **2** is oriented differently with respect to the arm **1**, typically when it pivots 180° in a vertical plane, the front connection component is the component **12B**, with the rear component being the component **12A**, provided that they are identical.

(19) The device **7** is equipped with at least two coupling components **13**, **14** with connection components **12A** and **12B**, respectively. By analogy, the component **13** will be called front coupling component and the component **14** will be called rear coupling component. The front coupling component **13**, located to the left of the part **11** with reference to FIG. 2, is configured as a hook, the opening **15** of which is oriented toward the side **16** defining a lateral edge of the device **7**, in this case the left edge with reference to FIG. 2 connecting the parts **9** and **11**. In other words, the opening **15** of the coupling component **13** is located as close as possible to the end **3** of the arm **1**. It is therefore oriented toward the cab of the public works machine, not shown. The opening **15** defines a housing for accommodating the connection component **12A**.

(20) The rear coupling component **14** is configured as a U-shape, the opening **17** of which is oriented toward the lower edge **18** of the device **7** with reference to FIG. 2. The opening **17** defines a housing for accommodating the connection component **12B**.

(21) As shown in FIG. 2A, a security component **19** is mounted on the part **11**, at the opening **15** of the coupling component **13**. It is shown in perspective in FIG. 11 and, laterally, alone and on a large scale, in FIG. 12. As can be particularly seen from FIGS. 2A, 11 and 12, the security component **19** is pivotably mounted in a plane **P** about a rotation shaft **20**, with a longitudinal geometric axis **A20**. The component **19** pivots in one direction or the other following the double arrow **F19**. The plane **P** is substantially perpendicular to a plane **P1** connecting the rotation shafts **6** and illustrated in FIG. 2. The attachment **7** in this case is equipped with a single security component **19**, as illustrated in FIG. 11. As an alternative embodiment, there are several components **19**, depending on the dimensions of the connection components **12A**, **12B**.

(22) The security component **19** is a part with a complex geometric shape, as shown in FIG. 11. It

has an elongated shape, with a complex cross-section. The dimensions, in particular the length, of the security component **19** are adapted to the dimensions of the connection component **12A**. As shown in FIG. **11**, the ends of the shaft **20** extend beyond the ends of the component **19** and ensure that the component **19** is fixed in the sides of the coupler **7**. For greater clarity, the sides of the coupler **7** are not shown in the various figures. It is easy to understand that the part **9** of the coupler **7** forms a protection for the active components of the part **11** of the coupler **7**. Hereafter, the active engaged position of the component **19** will be defined as a position in which it prevents the connection component **12A** from exiting its housing **15**. The disengaged position of the component **19** corresponds to a position in which it does not prevent the component **12A** from exiting its housing **15**.

(23) The security component **19** comprises, in the lower part with reference to FIGS. **11** and **12**, a part **21** with a round relief, which will be referred to as flap hereafter. This flap **21** has a radius of curvature that is adapted to the radius of the cylinder with a circular base forming the connection component **12A**.

(24) The shape of the curvature of the relief **21** is complex, as shown in FIG. **12**. Typically, the curved free edge **210** comprises a plurality of straight and curved surface segments. Each segment performs a given function, in relation to the position of the component **19**. A first straight and generally flat surface segment **211**, located to the right of FIG. **12**, is adapted to face, and if necessary come into abutment against, a portion of the connection component **12A** when the component **12A** is in place in the housing **15** of the coupling component **13**, as illustrated in FIG. **7**, with it being understood that the abutment only occurs if moving the component **12A** out of the housing **15** is initiated. In this case, the flap **21**, and in particular the segment **211**, secures the connection and prevents any uncoupling of the component **12A** and the coupler **7**. This position corresponds to a position, called active engaged position, of the component **19** and of the flap **21** when the connection component **12A** is installed in its housing **15**. In the working configuration, in which there is no movement of the component **12A** with a view to it exiting the housing **15**, a clearance of a few millimeters is provided between the segment **211** and the component **12A**. In other words, the flap **21** is in the active engaged position, ready to come into abutment against the component **12A** present in its housing **15**, but without being in abutment against the component **12A** so as not to generate any friction between the component **12A** and the component **19**.

(25) To the left of the segment **211**, the segment **212**, which is also straight and substantially flat, is arranged angularly. The segment **212** is a release segment, allowing any blocking contact between the flap **21** and the component **12A** to be avoided during the respective rotations of these two elements with respect to each other. To this end, the width of the segment **212** is the smallest of all the constituent segments of the edge **210**.

(26) To the left of the segment **212**, the segment **213** comprises a straight portion positioned between two curved portions. The shape of the segment **213** allows, when the component **12A** exits, the component **19** to rotate in the counterclockwise direction and therefore allows it to be placed in the disengaged position illustrated in FIG. **10**, so that the flap **21** does not prevent the connection component **12A** from exiting its housing **15**. The counterclockwise direction is indicated with reference to FIG. **12**, namely a rotation from right to left with reference to FIG. **12**. Thus, the counterclockwise direction is defined as being the rotation of the security component **19** toward the lateral edge **16** of the part **11** of the coupler **7**. In this position, the security component **19** can automatically return to a position, called reset position, as illustrated in FIG. **2**, **2A** or **3**.

(27) It should be noted that the security component **19** not only can be in a disengaged position where the security component **19** cannot prevent a movement involving the component **12A** exiting its accommodation housing **15** and in an active engaged position for blocking the component **12A** from exiting its housing **15**, as described above, but also in another position, called reset position, once the component **12A** is completely free of its housing **15** and is spaced apart from the component **19**. In this reset position, the security component **19** is remote from the component

12A, with the housing **15** being empty and ready to accommodate a component **12A**, with the security component **19** itself being ready to transition to the disengaged position, then to the active engaged position when the component **12A** is in place in the housing **15**, which is the case whereby the component **12A** is inserted into its housing **15**. In the reset position illustrated in FIGS. **2**, **2A** and **3**, the flap **21**, and therefore the component **19**, is in a position similar to that illustrated in FIGS. **5** to **8**, therefore a position corresponding to the engaged position. The component **19** can therefore occupy, in addition to a disengaged position, an engaged and active position, in which it blocks the component **12A** from exiting the housing **15**, and an engaged and inactive position where the housing **15** is empty and ready to accommodate a component **12A**. The implementation of the constituent segments of the free edge **210** of the flap **21** occurs during a translation movement of the component **12A**, in either direction, and induces, due to the configuration of the segments **211** to **213**, a rotation of the component **19**. The rotation of the component **19**, and therefore of the flap **21**, always occurs in the counterclockwise direction as defined above, therefore a rotation toward the edge **16** of the part **11** of the coupler **7**.

(28) As shown in FIGS. **1** to **10**, the dimensions of the flap **21** are such that it partially extends, in the active engaged position and in the inactive engaged position, called reset position, into the opening **15** defined between the coupling component **13**, configured as a hook forming one of the constituent jaws of the component **13**, and an edge **22** of the part **11** of the coupler **7** forming the other jaw. A part, not referenced, of the edge **22** is substantially parallel to the rectilinear part **130** of the hook **13** and thus defines a U-shape therewith. In this way, the U-shaped opening **15** forms a housing for accommodating the connection component **12A** of the tool **2**. As an alternative embodiment, the hook **13** defines a housing for accommodating the connection component **12B**, by turning over the tool **2** and provided that the components **12A** and **12B** are identical. In the reset position of the security component **19** illustrated in FIGS. **2** and **2A**, therefore in a position in which there is no coupling between the tool **2** and the quick coupler **7**, the flap **21** is in a position corresponding to the engaged position of the security component **19**, where its engagement in the opening **15** is maximal, with it being understood that the accommodation housing defined by the opening **15** is empty. Such a position of the flap **21** corresponds to an inactive engaged position, called reset position, identical to the engaged position of the flap **21** when the component **12A** is in place in its housing **15**. The return and the retention of the security component **19**, therefore of the flap **21**, in this position is achieved by at least one return component, in this case springs **23**, of the helical type and shown in FIG. **11**. As an alternative embodiment, the return to position is obtained using another means, known per se, for example, a leaf spring, a tension spring, a compression spring, a jack. By virtue of the return component of the security component **19**, the passage of said security component between the various positions—engaged and active with blocking of the component **12A** in its housing **15**, disengaged without blocking of the component **12A** in its housing **15**, engaged and inactive, called reset position, therefore in the absence of the component **12A** in its housing **15**—occurs without user intervention, only by the various movements of the component **12A** relative to its housing **15** and a release component, described hereafter. Such an operation, which is purely mechanical, allows optimum securing of the connection between the component **12A** and the coupler **7**, irrespective of the state of this connection.

(29) The security component **19** comprises, above the flap **21** with reference to FIGS. **2A** and **12**, a face **24** that is inclined toward the hook **13** and is flat for the most part. The high end **25** of the face **24**, therefore that furthest from the hook **13**, is curved and oriented inward, therefore toward the face **24**. Thus, as a cross section, the face **24** is generally in the shape of a crook. The end **25** is extended by a face **26** that is flat and generally orthogonal to a longitudinal axis **A20** of the rotation shaft **20**. A flat face **27**, generally perpendicular to the face **26**, connects said face **26** to the lower face **28** that supports the flap **21**.

(30) The part **11** accommodates at least one release component **29**. Said release component is positioned above the coupling components **13** and **14**, in the vicinity of the security component **19**.

The release component **29** comprises a finger **30** configured as a solid cylinder with a cylindrical base, the free end **31** of which generally assumes a shape complementing that of the curved end **25** of the security component **19**. Advantageously, the end of the finger **30** is configured as a cylinder with a circular base, the longitudinal axis of which is perpendicular to the longitudinal axis of the finger **30**. As an alternative embodiment, not shown, the finger **30** is flattened, in the shape of a connecting rod.

(31) The end **32** of the finger **30**, opposite the end **31**, is detachably or non-detachably inserted into a part forming the main body **33** of the release component **29**. The body **33** is, in the example, configured as a hollow cylinder with a circular base. As an alternative embodiment, it assumes another shape, for example, a parallelepiped. One end **34** of the body **33** receives the end **32** of the finger **30**. The opposite end **35**, referenced in FIG. **11**, of the body **33** is connected by a pivot link to a relief **36** forming a rotation support for the body **33**. The body **33** rotates in a plane parallel to the plane P of rotation of the security component **19**. The relief **36** is integrally formed with a lug **38** in the form of a hook and is slidably mounted in at least one slide **37**. The translation movement of the lug **38**, and therefore of the rotation support **36**, occurs in a plane orthogonal to the plane P1. The translation movement of the lug **38** in the slide **37** is implemented by a displacement means, typically by a jack, hydraulic or other. As an alternative embodiment, the translation movement is manual. In all cases, the movement of the lug **38** is not automatic but requires an action on the part of the user of the coupling assembly, therefore of the driver of the public works machine. The opening of the hook-shaped lug **38** is oriented toward the opening **17** of the first coupling component **14**. The lug **38** is dimensioned and positioned so that, during its translation movement in the slide **37**, following the double arrow F36, it releases or blocks access to the opening **17**. In other words, the lug or hook **38** allows the component **12B** to be blocked in position in the housing defined by the opening **17** of the coupling component **14**. De facto, the hook **38** grants or does not grant access allowing the connection component **12B** to enter or exit the housing **17** defined in the coupling component **14** according to the position of the hook **38**.

(32) The finger **30**, forming a first part of the release component **29**, is translationally movable in the body **33**, following the double arrow F30. The finger **30** is retained, in the inactive position, in a position where it extends as much as possible outside the body **33** using a return component or pusher **39**, as illustrated in FIGS. **2** and **2A** in particular. The return component **39** in this case is a mechanical spring jack. As an alternative embodiment, it is another type of return component known per se, for example, a pneumatic or hydraulic jack. In all cases, the return component **39** acts in order to permanently push the finger **30** out of the body **33**, irrespective of the position of the finger and irrespective of the forces acting on the finger.

(33) The body **33**, forming a second part of the release component **29**, is provided with a hook **40**, the opening of which is directed toward the finger **30**. This hook **40** is adapted to detachably accommodate one end **41** of a helical spring **42**, the other end **43** of which is connected to a hook **44** that is fixed on a face of the lug **38**. When the spring **42** is connected to the two hooks **40** and **44**, its return force induces a pivoting movement of the assembly of the release component **29** oriented downward following the double arrow F29, this occurs by rotating the end **35** of the body **33** about the rotation shaft **45** equipping the rotation support **36**. In another embodiment, the helical spring **42** is replaced by another return component, for example, a leaf pressure spring positioned above the body **33** and exerting a pressure on the body **33** and inducing a pivoting movement of the whole of the release component **29** that is also oriented downward. In all cases, using the spring **42** or using another return component, the release component **29** is retained in a position such that it is always in contact with or facing, at least by one zone, in particular its free end **31**, the security component **19**, in particular with the flap **21**, with this occurring during the various translation and rotation movements of the components **19** and **29**. Thus, the release component **29** is facing the security component **19** when said security component is in the active engaged position and is ready to ensure, through contact, that the component **12A** is blocked in the housing **15** when an attempt is

made to remove the component **12A** from the housing.

(34) The operation of the secure connection device or quick coupler **7** will now be described with reference to the various FIGS. **1** to **14**. The operation will first be described with reference to attaching a tool **2** on the device **7**, therefore when installing the tool **2** on the arm **1** of a public works machine, then when removing the tool **2**, for example, during a tool change.

(35) In all cases, when attaching the tool, this necessarily implies that the tool is stationary relative to the arm **1**, for example, placed on any surface, generally on the ground. In one embodiment, not shown, a security device can be provided that comprises sensors placed on the quick coupler **7** and/or the arm **1** and/or the tool **2** indicating, using an audible and/or visual signal, that the components **12A**, **12B** are engaged in the connection components **13**, **14** and that the coupling between the tool **2** and the coupler **7** is complete and secured.

(36) FIGS. **2** and **2A** illustrate a configuration in which the tool **2** and the quick coupler **7** are completely separated, for example, before mounting or after dismounting the tool **2** on/from the arm **1** of a public works machine, therefore a configuration in which all the constituent elements of the coupling assembly are inactive. In this case, the openings or housings **15** and **17** of the coupling components **13** and **14** are completely free and accessible. The security component **19** is in the position in which the flap **21** extends as much as possible into the housing **15**, which is empty. This corresponds to an engaged and inactive position of the security component **19**. The face **27** of the security component **19** is substantially vertical and the face **26** is substantially horizontal. The flap **21** is retained in this position by the return of the spring **23**. The finger **30** of the release component **29** is in abutment on the face **26** of the security component **19**. In this configuration, the free end **31** of the finger **30** is at its top dead center or at the end of its travel, above the locking component **19**. In this position, the finger **30** and the body **33** of the release component **29** are inclined upward, with reference to FIGS. **2** and **2A**, by pivoting the body **33** about the shaft **45**. The return force of the spring **42** tends to bring the release component **29** downward, therefore to keep it in abutment against the face **26**. In this case, the contact zone between the security component **19** and the finger **30** is formed, respectively, by a portion of the face **26** and a portion of the outer surface of the finger **30**, which in this case is cylindrical with a circular base. The security component **19** is ready to pivot, with the release component **29** not opposing a movement of the security component **19** when the component **12A** is inserted into its housing **15**. The component **19** then pivots in the counterclockwise direction, with reference to FIGS. **2** to **10**, therefore toward the edge **16** of the part **11** of the coupler **7**. Such a direction of rotation corresponds to a rotation of the component **19** toward the lower edge **22** of the part **7**, tending to bring the edge **27** of the component **19** parallel to the edge **22**. In other words, the end **26** of the component **19** pivots toward the opening **15**.

(37) In this configuration, the hook-shaped lug **38** is at an end of travel dead center, corresponding to free and maximum access to the opening **17** of the first coupling component **14**. In other words, the locking component formed by the lug **38** is in the inactive position. The lug **38** is retained in this position by the fact that the jack, which is schematically shown by reference sign **370**, which activates the lug **38** and therefore the release component **29**, is inactive in the retracted position. Starting and stopping the jack **370** are controlled by the user of the coupling assembly, for example, the driver of the public works machine from the operating cab or by a user equipped with a remote control unit and remote from the machine, for example, when said machine is automated and does not comprise an operating cab.

(38) FIG. **3** illustrates an initial phase of positioning the quick coupler **7** in order to connect with the tool **2**, with said tool being stationary on a surface, the connection components **12A**, **12B** being accessible, the edge **200** of the tool **2** being oriented upward. Firstly, the coupling component **13** needs to be arranged facing the connection component **12A**. The passage of the component **12A** in the opening **15** of the coupling component **13**, toward the round bottom **131** of the component **13**, following the double arrow **F13**, induces contact between the flap **21**, therefore the security component, and the periphery of the body **12A**. The contact is initially made by the segment **213** of

the flap **21** and the component **12A**. The round relief **21** has a radius of curvature of the edge **210** that is adapted to that of the cylinder with a circular base forming the component **12A**, so that the contact between these two parts is of the sliding type. In other words, such a type of contact allows the parts to move relative to each other. Thus, the translation movement of the component **12A** toward the bottom **131** of the coupling component **13** rotates the security component **19**, by means of the pivoting of the flap **21**, with the segments **213** or **212** of the edge **210** being in contact with the component **12A**. The security component **19** then pivots in the counterclockwise direction, as previously defined following the double arrow **F19**. This pivoting brings the flap **21**, and therefore the security component **19**, into a disengaged position, illustrated in FIG. 4, in which it is generally aligned with the edge **22**. In this position, the flap **21** does not extend into the housing **15** of the component **13**. In the embodiment illustrated in FIG. 4, the pivoting of the flap **21** in this direction is limited by a stop **190**.

(39) In this disengaged position of the flap **21**, and therefore of the component **19**, induced by the component **12A** when it is inserted into the housing **15** defined by the coupling component **13**, the passage of the component **12A** is free and unimpeded. In addition to suppressing the flap **21**, pivoting the component **19** brings the face **26** to a bottom dead center. In this position, the finger **30** no longer rests on the face **26**, as illustrated in FIGS. 2 and 3, but only on a corner **46** defined by the junction between the face **26** and the curved end **25** of the inclined face **24**. The contact surface between the security component **19** and the release component **29** is therefore minimal. The return force exerted by the spring **42** constantly keeps the finger **30** in abutment on the component **19**. The finger **30**, and in particular its free end **31**, has therefore lowered and is located in the vicinity of the round part **25** of the security component **19**.

(40) The movement of the component **12A** toward the round bottom **131** of the component **13**, following the double arrow **F13**, continues until the component **12A** is in abutment on the bottom **131** and on the rectilinear part **130** of the component **13**. The component **13** then surrounds the component **12A** by more than two thirds of its circumference, as illustrated in FIG. 5. In this configuration for retaining the component **12A** in the component **13**, the flap **21** has returned to a position similar to its initial position, which is the inactive engaged position initially illustrated in FIG. 2, with the component **12A** not being present in FIG. 2. This return to position occurs automatically, by pivoting the component **19** in the other direction following the double arrow **F19**. This is rendered possible by the fact that the segments **212** and **213** of the flap **21** are in sliding contact with peripheral complementary shaped surfaces of the component **12A**, the flap **21** and the component **12A** with appropriate radii of curvature. In order for the rotation of the flap **21**, and therefore of the security component **19**, to occur automatically in the opposite direction to the rotation of the component **12A**, with the finger **30** being in abutment on the component **19**, there is a provision for the return force of the spring **23** to be greater than the return force of the spring **42**.

(41) In the position illustrated in FIG. 5, the free end **31** of the finger **30** is substantially located in the same position as that illustrated in FIG. 3. FIG. 5 therefore represents the retention of the component **12A** in the component **13** and the securing of this retention by the flap **21**. At this stage, the component **12B** is not in position in the component **14** but it is facing the opening or housing **17** of the component **14**. To introduce the component **12B** into the component **14**, the coupler **7** needs to be lowered toward the tool **2**, following the double arrow **F7**. It should be noted that, at this stage, the coupling between the components **12A** and **13** is secure, any accidental removal of the component **12A** from the housing defined by the component **13** is impossible due to the engaged position of the security component **19**, namely the position where the segment **211** of the flap **21** is facing the component **12A** and prevents the component **12A** from exiting the housing **15**. This configuration is achieved mechanically, irrespective of the position of the tool **2** and of the coupler **7**, without any user intervention on the security component **19**. In one embodiment, a visual and/or audible signal, for example, based on sensors on the component **12A**, and/or the segment **211**, and/or the component **13**, notifies the user of the secure coupling between the components **12A** and

13. In other words, at this stage, even if the tool swings, said tool remains attached to the coupler **7**, without any risk of falling.

(42) The coupler **7** is then positioned as shown in FIG. **6**. The two other components **12B** and **14** now need to be coupled, by pivoting the coupler **7** about the component **12A** following the double arrow F7. At the end of the movement, the connection component **12B** is in abutment on the bottom **140** of the coupling component **14**. In this configuration, the coupling of the components **12B** and **14** is not secure, with the component **12B** being able to freely exit the component **14**, by a pivoting movement in the opposite direction to that previously described following the double arrow F7. In other words, in this position the coupling between the tool **2** and the coupler **7** is not completely achieved, with only the connection between the component **12A** and the component **13** being secure. It is possible to provide, in another embodiment, an audible and/or light signal notifying the user that the coupling is not yet complete and secure. To this end, for example, contactors are available on the bottoms **131**, **140** of the components **13** and **14**.

(43) To secure the connection between the component **12B** and the component **14**, the hook-shaped lug **38** must translationally move following the double arrow F36 toward the component **14**. This movement is performed by sliding in the slide **37**. The sliding movement is advantageously generated by the jack **370** controlled by the user, either from the operating cab of the public works machine or by a remote control. As an alternative embodiment, the movement of the hook or lug **38** is performed by a manual action, therefore with the intervention of the user, using a device for pushing or pulling the hook **38**. When the hook or lug **38**, which defines a locking component, is in abutment against the component **12B**, therefore in the active position, it locks the retention of the connection component **12B** by the coupling component **14**. The hook shape of the lug **38** prevents any removal of the component **12B**. In the position illustrated in FIG. **7**, the security component **19** is in the active engaged position and secures the retention of the component **12A**. The retention of the component **12A** by the security component **19** is secured before transitioning to the active position of the locking component **38**, which locks the retention of the component **12B**. Locking the retention of the components **12A** and **12B** occurs one after the other. The tool **2** is thus securely coupled with the quick coupler **7**.

(44) In this configuration, the release component **29** has moved back following the double arrow F36, to a second end of translation movement travel limit. This movement has moved, by sliding the finger **30** on the face **26**, the finger **30** substantially opposite the round part **25** but without any contact with the part **25** or with the face **24** of the security component **19**. This movement is provided by the return force exerted by the spring **42** on the body **33**. This movement, which is directed downward following the double arrow F29, is limited by the support **36**, which, configured as a U-shape, defines a stop that stops any pivoting of the body **33** and therefore of the finger **30**. In this configuration, the tool **2** can be used without any risk of accidental unhooking thereof. In another embodiment, an audible and/or light signal notifies the user of the complete and secure coupling of the tool **2** on the coupler **7**. Such light signals can be obtained by color markings of the parts or parts of parts forming the coupling assembly. As an alternative embodiment, sensors fixed on the various connection and coupling components are used and generate an audible and/or light signal where the user is located, generally the cab of the public works machine.

(45) To unhook the tool **2** from the coupler **7**, the movement illustrated in FIG. **8** is initiated. To this end, the tool **2** needs to be maintained a few centimeters or decimeters above a surface, in order to allow the component **12B** to disengage from the housing defined by the coupling component **14** under the effect of gravity, by pivoting the tool **2** about the connection component **12A**. Once the coupling component **12B** has been disengaged, the tool is placed and immobilized on a surface in order to continue uncoupling the tool **2** and the coupler **7**.

(46) This movement involves moving the locking component **38** in the direction of the double arrow F36 but in the opposite direction with respect to the movement that allowed coupling between the components **12B** and **14**. In other words, in this case, the movement aims to move the

hook **38** away from the component **12B**. This movement brings the end **31** of the finger **30** into contact against the round part **25** of the security component **19**. The spring **42** retains the body **33** and therefore the finger **30** in a horizontal position, with the body **33** remaining in abutment against the support **36**.

(47) By continuing the movement of the hook **38** in the slide **37** to the end of the stroke of the jack **370** or of the manual translation means, the release component **29** is pushed back toward the component **13**, as illustrated in FIG. **9**. Since the body **33** is pivotably mounted about the shaft **45** and due to the complementary shapes of the end **31** and of the face **25**, the movement of the finger **30**, when continued beyond the simple contact between the face **25** and the end **31**, is no longer horizontal, but tends to be slightly curvilinear and inclined upward, following the arrow **F31** in FIG. **9**. In this way, the translation movement of the hook **38** thrusts the finger **30** on the face **25** and initiates the rotation of the locking component **19**, and therefore of the flap **21**, about the axis of rotation **A20**, following the double arrow **F19**. The rotation occurs in the same counterclockwise direction as described above, when the component **12A** is introduced into its housing **15**. This movement brings the flap **21** opposite the component **12A**. In this position, the flap **21** is in a generally completely disengaged position, therefore substantially coplanar with the edge **22** and it no longer opposes the removal of the component **12A**, following the double arrow **F31**. To this end, the attachment **7** is translationally moved by a translation movement in the opposite direction to that described in FIGS. **3** to **5**, in order to uncouple the components **12A** and **13**. The component **12A** is removed with constant contact with the flap **21**, in particular with the segment **212** of the flap **21**.

(48) By continuing to remove the component **12A** from the component **13**, as shown in FIG. **10**, the tool **2** is completely released from the secure connection device **7**. When the component **12A** is removed, a sliding contact is made between the segment **213** of the flap **21** and the component **12A**. During this contact, due to the configuration of the segment **213** and its bearing on the component **12A**, the flap **21**, and therefore the security component **19**, is rotated during the translation movement of the component **12A**. The security component **19** therefore rotates in the counterclockwise direction, as described above, and continues until the end **31** of the finger **30** passes above the component **19** and only rests on the security component **19** at the corner **46**, in the same position as that illustrated in FIG. **4**. A hard stop **190**, or, as an alternative embodiment, an adjustable screw introduced into part of the wall **22** of the coupler **7**, limits the rotation of the security component **19**.

(49) This removal movement of the component **12A** therefore allows the security component **19** to return to the reset position, with the flap **21** and the surfaces **211** to **213** then being in the inactive engaged position, so that the component **19** is ready to be implemented for the following connection between a tool **2** and the coupler **7**, without any human intervention, mechanically and automatically. Indeed, when the component **12A** is completely disengaged from the component **13**, nothing opposes the return force of the spring **23** and the security component **19** returns to its inactive engaged position and the flap **21**, and therefore the surfaces **211** to **213**, returns to the inactive engaged position, as in FIG. **2A**. It is then possible to reset a procedure for coupling a tool on the quick coupler.

(50) FIGS. **13** and **14** illustrate another embodiment of the invention. Common elements with the previous embodiment have the same reference signs. The distinct elements of this embodiment use the same reference signs of the elements with the same function of the previous embodiment, multiplied by 10. A finger **300**, which can have a circular, square, rectangular or oval cross-section, is inserted via one end into a body **330** that is fixed directly on the lug **38**. In this way, only a translation movement of the finger **300**, in the same direction as the movement of the lug **38**, is possible. The length of the finger **300** is such that, in the configuration illustrated in FIG. **13**, in which the coupling between the components **12A**, **12B**, **13** and **14** is locked and secure, it occupies almost all the space available in the coupler **7**. A spring, similar to the spring **39**, is inserted into the

body **330** and ensures that a constant thrust force is applied to the finger **300**. The finger **300** is provided with a notch **301**. The notch **301** has a generally inclined bottom, toward the body **330**, with a substantially vertical wall defining the edge of the notch **301** closest to the body **330**. A pawl **302**, pivotably mounted on the component **19**, is installed in the notch **301**. The shape and dimensions of the pawl **302** are adapted to those of the notch **301**, with said notch defining a housing for accommodating the pawl **302**. The pawl **302** is mounted on a position return spring, not shown. When uncoupling is initiated, similarly to the first embodiment, pivoting of the component **19** is initiated, by means of the pawl **302** engaged in the notch **301**, subject to the thrust of the finger **300**, which moves translationally. In this case, pivoting the component **19** brings the pawl in the vicinity of a pin **303** secured to one of the walls of the coupler **7**. The pin **303** has a substantially circular section in the example. In all cases, the shape and the position of the pin **303** are such that when there is contact between the pin **303** and the pawl **302**, this causes said pawl to tilt, which releases it from the notch **301**. The translation movement of the finger **300** continues and the rotation movement of the component **19** continues until the disengaged position is reached, illustrated in FIG. **14**, which allows the component **12A** to exit the housing **15**. The exit of the component **12A** causes an additional rotation of the component **19**, which causes the continuation of the movement of the pawl **302** by the pin **303**. The pawl **302** is then disengaged from the notch **301**, which allows the component **19** to freely rotate and therefore to return to the reset position, in a position defined by a stop, not shown, with the component **12A** being out of its housing **15**. During this movement, according to an advantageous embodiment, the free end **304** of the finger **300** extends beyond the coupler **7** through an orifice provided therein. The end **304** defines a visual signal for the user, as illustrated in FIG. **14**, once the tool **2** is released from the coupler **7**. Such a signal allows the state of the device to be checked, namely locked or unlocked depending on whether or not the finger extends out of the coupler **7**. In another embodiment, not illustrated, the pawl is fixed on the finger **300**, with a notch being provided on the component **19**. In this case, the pin **303** is fixed on the component **19**, causing the pawl to tilt out of the notch. In the embodiment of FIGS. **13** and **14**, the maintenance of the device is facilitated, by limiting any ingress of dust and dirt into the coupler **7**. Indeed, the presence of a single translation movement of the release component allows a seal to be used at the various openings, both in the body **330** and the opening provided in the face of the coupler **7** through which the end **304** of the finger **300** exits. In addition, the risk of an operating incident is also limited, with the kinematics being simplified.

(51) By virtue of the invention, no human intervention is necessary for the security component to return to the reset position. This resetting is completely mechanical, as is the blocking of the security component **19** in the engaged position by the respective movements of the connection and coupling components and while maintaining kinematics for implementing the coupler **7** that are similar to those of the prior art. The invention offers, unlike the prior art, ternary operation of the security component, with automatic passage, without human intervention and by mechanical means, between three noteworthy positions: an active engaged position of the component **19**, in which it blocks the component **12A** from exiting its housing, a disengaged position of the component **19**, in which the component **12A** can exit its housing, and an inactive engaged position, called reset position, of the component **19**, in which the component **19** has no effect on the component **12A**, with said component being remote from the component **19**, which, in the engaged position, is ready to repeat a connection cycle with the component **12A**.

(52) The transition between these three positions is induced by the movements of the component **12A** relative to its housing **15** and by the movement of the unlocking component **29**, when the component **12A** enters or exits its housing **15**. Such kinematics are obtained by a particular geometric configuration of the security component **19** and of the unlocking component **29**, of the return components **42**, **370**, as well as by their respective positions in the coupler **7**.

(53) As a result, no human intervention is necessary for resetting the security component, since this occurs automatically using mechanical means, without any risk of forgetting or of errors. In other

words, this optimizes the safety of the operations for assembling and disassembling a tool on an arm of a public works machine, by avoiding any malfunctions of a hydraulic or electrical system or any omission or incorrect operation by the user. Furthermore, maintaining such a coupling assembly is optimized, since the constituent elements are easy to dismantle.

(54) In an advantageous embodiment, the slides **37** are detachably mounted in the coupler **7**, facilitating the cleaning and maintenance of the coupler.

(55) In one embodiment, the user is visually and/or audibly notified of the resetting of the safety device **19**, or more generally its position.

Claims

1. An assembly for coupling a tool (**2**) on a working arm (**1**) of a public works machine, comprising: a secure connection device (**7**), called a quick coupler, two parallel connection components (**12A**, **12B**) equipping the tool (**2**), at least one connection component of the two parallel connection components (**12A**, **12B**) being a cylinder with a circular base, two coupling components (**13**, **14**) configured to equip the secure connection device via the two connection components (**12A**, **12B**), the two coupling components (**13**, **14**) defining housings (**15**, **17**) for accommodating the two parallel connection components (**12A**, **12B**), a security component (**19**) mounted to freely rotate (**F19**) between a disengaged position, in which the security component allows a first connection component (**12A**) of the two parallel connection components equipping the tool (**2**) to move relative to a first coupling component (**13**) of the two coupling components, an engaged position, in which the security component secures retention of the first connection component (**12A**) in the first coupling component (**13**) and a reset position, in which the first connection component (**12A**) is remote from the first coupling component (**13**), without any coupling between the tool (**2**) and the connection device (**7**), at least one locking component (**36**, **38**) that is movable (**F36**) between an inactive position, in which retention by a second coupling component (**14**) of the two coupling components equipping the secure connection device (**7**) of a second connection component (**12B**) of the two parallel connection components is not locked, and an active position, in which the at least one locking component locks the retention by the second coupling component (**14**) of the second connection component (**12B**), wherein a force acting on the at least one locking component (**36**, **38**) moves the at least one locking component from the active position to the inactive position and from the inactive position to the active position, a release component (**29**) that is adapted to allow the security component (**19**) to rotate (**F19**) from the engaged position to the disengaged position in response to movement (**F36**) of the at least one locking component (**36**, **38**), the release component (**29**) being pivotably secured (**33**) on the at least one locking component (**36**, **38**) and comprising a finger (**30**) that is translationally movable (**F30**), the finger (**30**; **300**) of the release component (**29**) adapted (**30**, **301**) to be in contact with at least one release component surface (**26**, **46**, **25**, **24**; **302**) of the security component (**19**), and the security component (**19**) further comprising at least one first connection component surface (**21**, **211**, **212**, **213**) adapted to be in sliding contact with the first connection component (**12A**), the sliding contact of the at least one first connection component surface with the first connection component (**12A**) inducing a rotation (**F19**) of the security component (**19**) from the reset position to the disengage position when the first connection component (**12A**) moves into its accommodation housing (**15**) defined by the first coupling component (**13**) while the at least one locking component (**36**, **38**) is in the inactive position, wherein the rotation (**F19**) of the security component (**19**) to allow the first connection component (**12A**) to move out of the accommodation housing (**15**) defined by the first coupling component (**13**) occurs in a counterclockwise direction due to the at least one locking component (**36**, **38**) being moved from the active position to the inactive position, causing the finger to contact the at least one release component surface of the security component, the finger forcing the securing component to rotate toward an edge (**22**) of a

- lower part (11) of the connection device (7) and moving the security component from the engaged position to the disengaged position, and after removal of the first connection component (12A) from the accommodation housing, the release component (29) allows the security component (19) to return to the reset position.
2. The coupling assembly according to claim 1, wherein the security component (19) comprises surface a round relief or flap having the at least one first connection component surface (21), a radius of curvature of the round relief or flap adapted to radius of the cylinder with a circular base forming the first connection component (12A), the at least one first connection component surface (21) comprising at least three flat and curved surface segments (211, 212, 213) adapted to be successively at least facing at least part of a surface of the connection component (12A).
 3. The coupling assembly according to claim 2, wherein a segment (211) of the round relief (21) is permanently at least facing part of the surface of the first connection component (12A) when the first connection component is in the accommodation housing (15) defined by the first coupling component (13).
 4. The coupling assembly according to claim 1, wherein the security component (19) is retained in the engaged position by a return component (23).
 5. The coupling assembly according to claim 1, wherein the release component (29) comprises a main body (33; 330) configured to receive the finger (30; 300) that is translationally movable (F30) in the body (33; 330).
 6. The coupling assembly according to claim 5, wherein a return component (39) is inserted into the body (330) and ensures that a constant force is applied to the finger (300), retaining the finger (30; 300) outside the body (33; 330) of the release component (29).
 7. The coupling assembly according to claim 5, wherein a return component (42) for positioning the body (33) retains the body in a position in which the finger (30) is capable of interacting with the security component (19).
 8. The coupling assembly according to claim 5, wherein the finger (30) comprises at least one zone in contact with the security component (19) and located at a free end (31) of the finger (30) when the retention of the second connection component (12B) by the second coupling component (14) is not locked.
 9. The coupling assembly according to claim 5, wherein the finger (300) comprises a notch (301) adapted to receive a pawl (302) that is free to rotate relative to the security component (19).
 10. The coupling assembly according to claim 9, wherein the pawl (302) is secured to the security component (19), with a pin (303) for rotating the pawl (302) being secured to a wall of the quick coupler (7).
 11. The coupling assembly according to claim 9, wherein an end (304) of the finger (300) extends beyond the quick coupler (7) when the security component (19) is in the disengaged position, forming a visual signal for the user.
 12. The coupling assembly according to claim 1, wherein the locking component (36, 38) comprises a lug configured as a hook (38) that is translationally movable (F36) in slides (37).
 13. The coupling assembly according to claim 12, wherein the slides (37) are detachably mounted in the quick coupler (7).
 14. A public works machine provided with at least one working arm (1) equipped with the coupling assembly (7) for coupling the tool (2) on the working arm (1) in accordance with claim 1.
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